INTELLIGENT VISION-BASED MANUAL/AUTOMATIC ROBOTICS MANIPULATOR

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Introduction

1. Landmines remain a severe hazard in many regions, making conventional detection methods dangerous, time-consuming, and heavily dependent on human effort.
2. To eliminates human risk by using an advanced robot to detect metallic objects autonomously or via remote control.
3. Dual microcontrollers, live visuals, and automated marking make this landmine clearance system faster and safer..

Project Objectives

1. Develop a reliable and low-cost robotic platform for landmine detection
2. A dual-microcontroller architecture efficiently balances web hosting, video streaming, and hardware control.
3. Feature Dual-Mode Camera-assisted Manual and automatic operation.
4. Achieve real-time coordination to stop movement and activate the manipulator arm.

Proposed Method

The system consists of both hardware and software.

System Architecture

- Distributed processing for efficiency and stability.
- **ESP32-S3 CAM:** Handles web dashboard, user input, and live video streaming.
- **ESP32 DevKit:** Controls sensors, manipulator arm, and real-time tasks via FreeRTOS.
- **UART Bridge:** Transfers commands and alerts between both controllers.
- **L298N Driver:** Provides precise movement and turning control

Metal Detection Process

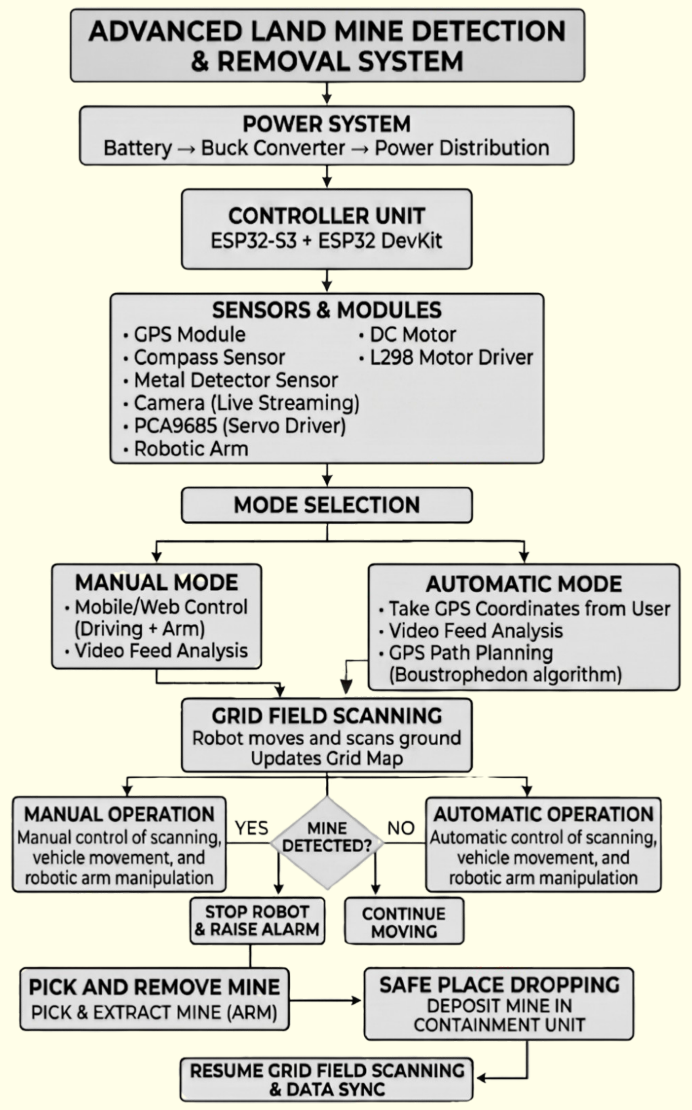
- Metal detector scans ground continuously.
- ESP32 processes detection instantly.
- L298N stops robot immediately.

Manipulator Activation

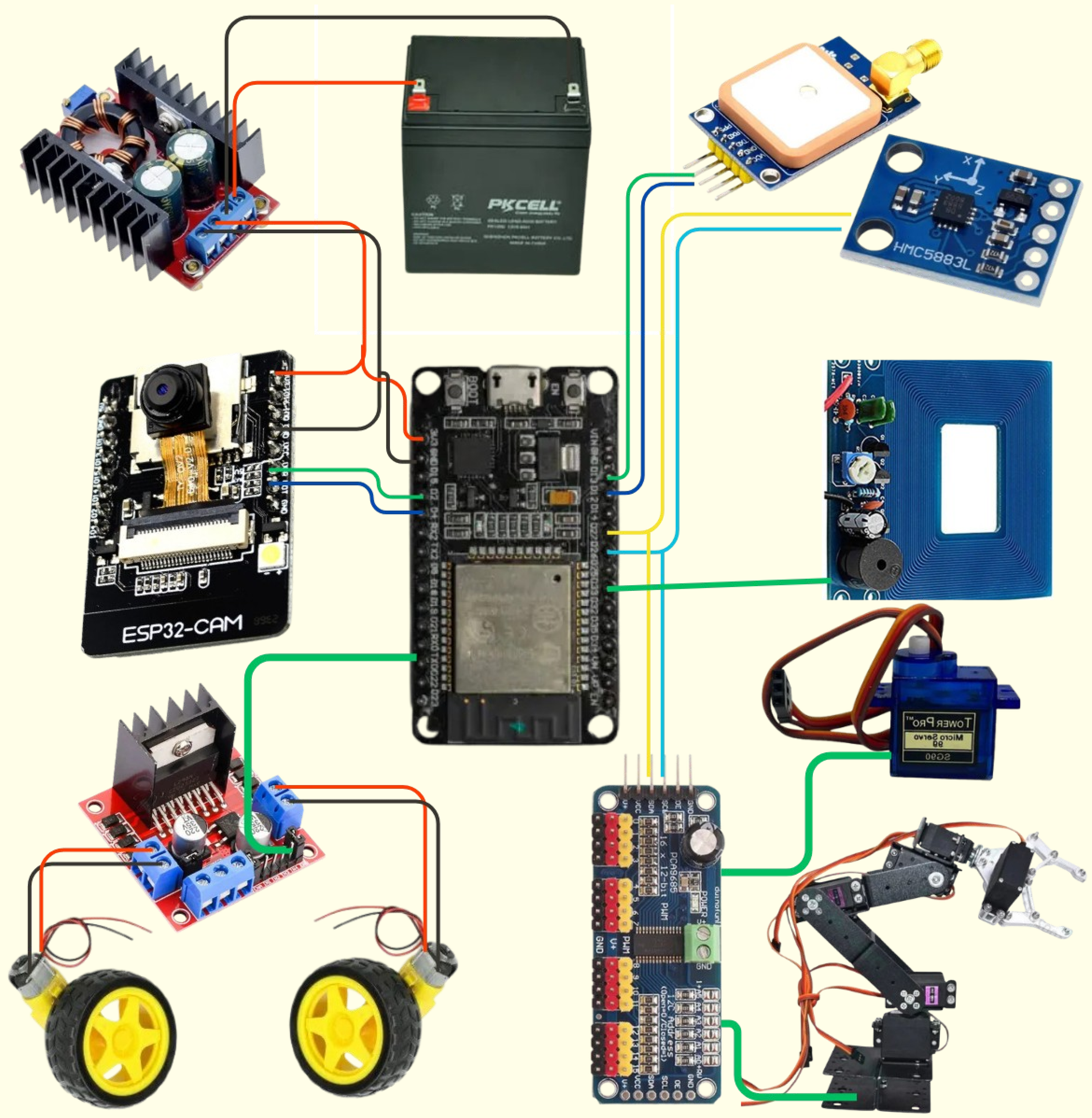
- **Automatic Action:** Arm activates after stop to mark or inspect target.
- **Manual Trigger:** Operator controls arm via dashboard using live camera feed.

Robot Operation Workflow

- **Manual Mode:** Dashboard controls robot live
- **Automatic Mode:** GPS guides autonomous navigation.
- **Continuous Scanning:** Metal detector runs always.



Systematic Diagram



Project Cost

Description	Cost in (PKR)
System Architecture (Sensors, Power Supply)	20,000/-
Control & Decision Logic (ESP32)	3000/-
Robotics Manipulator	15,000/-
Mobile Platform	7000/-
Miscellaneous (Wires, Soldering, Connectors, Housing, etc.)	5000/-
Total	**50,000/-**

Experimental Results and Discussion

- Dual controllers ensured stable video and fast control.
- UART bridge proved reliable for seamless data exchange between the web server and the hardware control nodeRobot accurately detected metallic objects.
- Emergency stop reacted instantly on detection.
- The robotic arm flawlessly executed its sequence, confirming perfect sensor-actuator coordination

Conclusion and Perspectives

Conclusion

1. The developed dual-ESP system successfully bridges the gap between low-cost hardware and complex safety logic.
2. By providing both camera-guided Manual and GPS-guided Automatic modes, the robot offers immense flexibility for different operational terrains.
3. This project drastically enhances safety and reduces human involvement in preliminary landmine detection tasks.

Future Direction

1. Integrate OpenCV with ESP32-S3 for visual threat detection.
2. Enhancing autonomous navigation accuracy using RTK (Real-Time Kinematic) GPS systems.
3. Upgrade sensors for deeper soil detection.

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